NET_SCAN

Automated Lateral Movement Detection & Cyber Kill Chain Modeling

SECURITY ASSESSMENT REPORT

Target: 8.8.8.8

Date: April 06, 2026 at 18:08:33

Ports Scanned: 1024

Open Ports Discovered: 3

RISK: HIGH

CONFIDENTIAL - For authorized personnel only

Generated by NET_SCAN Security Intelligence Platform

1. Executive Summary

This report presents the findings of an automated security assessment conducted against the target system at IP address 8.8.8.8. The assessment was performed using NET_SCAN's multi-threaded TCP port enumeration engine, followed by an automated intelligence pipeline that includes service identification, vulnerability correlation, kill chain classification, and lateral movement path analysis.

The scan evaluated 1024 TCP ports and discovered 3 open port(s) exposing network services. Each discovered service was analyzed for potential vulnerabilities and mapped to MITRE ATT&CK techniques to identify realistic attack vectors.

1.1 Key Metrics

Target IP Address	8.8.8.8
Date of Assessment	2026-04-06 18:08:33
Total Ports Scanned	1024
Open Ports Found	3
Closed Ports	1021
Filtered Ports	0
Total Vulnerabilities	2
Overall Risk Level	**HIGH**

1.2 Vulnerability Breakdown

Critical: 0

High: 3

Medium: 0

2. Exposed Services Matrix

The following table lists all discovered open ports with their associated services, risk classifications, and identified vulnerabilities. Services are classified based on their potential role in an attack chain (Entry, Pivot, or Target).

PORT	SERVICE	RISK	ROLE	VULNERABILITIES
53	DNS	HIGH	TARGET	Zone transfer possible, DDoS amplification
443	F	HIGH	ENTRY	None identified
853	Unknown	HIGH	-	None identified

2.1 Detailed Service Analysis

Port 53 - DNS

Risk Level: **HIGH**

Vulnerabilities:

- Zone transfer possible
- DDoS amplification

Port 443 - F

Risk Level: **HIGH**

No specific vulnerabilities identified in offline database.

Port 853 - Unknown

Risk Level: **HIGH**

No specific vulnerabilities identified in offline database.

3. Kill Chain & Attack Path Analysis

NET_SCAN classifies discovered services into kill chain roles and maps realistic lateral movement paths between them. The analysis uses a heuristic classification engine that assigns each service a role based on its typical use in adversarial attack campaigns:

ENTRY: Initial access vectors - services commonly exploited for first foothold (HTTP, FTP, RDP, VNC)

PIVOT: Lateral movement tools - services used to traverse internal networks (SSH, SMB, NetBIOS)

TARGET: High-value objectives - databases and services containing sensitive data (MySQL, MSSQL, DNS)

3.1 Node Classification Summary

Entry Points: 1 services

Pivot Nodes: 0 services

Target Nodes: 1 services

Total Chains: 1 attack paths identified

3.2 Identified Attack Chains

#	FROM	TO	RISK	MITRE ID	TECHNIQUE
1	Port 443 (F)	Port 53 (DNS)	8.9	T1210	Exploitation of Remote Service

3.3 Attack Chain Descriptions

Chain 1: Port 443 (F) -> Port 53 (DNS)

Attacker leverages F to compromise DNS

MITR

3. Threat Intelligence & Vulnerability Assessment

This section details the specific vulnerabilities identified for each open port based on NET_SCAN's offline vulnerability database. These findings should be cross-referenced with the NIST National Vulnerability Database (NVD) for the latest CVE data.

Port 53 (DNS)

- Zone transfer possible
- DDoS amplification

Impact: DNS poisoning, zone transfer data leakage, DDoS amplification

4. Network Exposure Score (NES) Methodology

The Network Exposure Score is a composite metric (0-100) that quantifies the overall risk posture of the target system. It considers multiple independent factors aligned with CVSS v3.1 scoring matrices and lateral movement compounding logic.

4.1 Scoring Factors

Open Ports with Critical CVEs

Ports hosting services with known critical vulnerabilities in NVD receive the highest weight.

Lateral Movement Path Count

The number of viable attack chains discovered between entry, pivot, and target nodes.

Service Vulnerability Severity

Individual risk levels of each discovered service based on common exploitation patterns.

Blast Radius Coverage %

Percentage of the network reachable from each compromised node through lateral movement.

Network Topology Depth

The depth of the attack graph - deeper graphs indicate more complex multi-hop attack opportunities.

4.2 Score Ranges

0 - 30	LOW	Minimal attack surface. Few or no exploitable paths detected.
31 - 60	MEDIUM	Moderate exposure. Some lateral movement paths exist but are limited.
61 - 80	HIGH	Significant exposure. Multiple attack chains with viable exploitation paths.
81 - 100	CRITICAL	Severe exposure. Extensive lateral movement with critical service compromise.

5. Mitigation Recommendations

Based on the assessment findings, the following remediation actions are recommended in order of priority. Implementing these measures will significantly reduce the attack surface and minimize the risk of lateral movement within the network.

1. **Implement Network Segmentation** — CRITICAL

   Isolate critical servers (databases, domain controllers) in separate VLANs with strict firewall rules. Prevent direct entry-to-target paths by enforcing traffic inspection between segments.

2. **Disable Unnecessary Services** — CRITICAL

   Close all non-essential ports, especially legacy protocols like Telnet (23), FTP (21), and unencrypted services. Replace with secure alternatives (SSH, SFTP, HTTPS).

3. **Apply Least Privilege Access** — HIGH

   Restrict service accounts to minimum required permissions. Disable default credentials on all discovered services. Implement role-based access control (RBAC).

4. **Deploy Intrusion Detection Systems** — HIGH

   Install network-based IDS/IPS at segment boundaries to detect lateral movement attempts. Configure alerts for MITRE ATT&CK techniques identified in this assessment.

5. **Implement Zero Trust Architecture** — HIGH

   Authenticate and encrypt all internal traffic. Do not trust any connection by default regardless of network location. Implement micro-segmentation.

6. **Patch Management Program** — CRITICAL

   Establish automated patch management for all exposed services. Prioritize patching based on CVSS scores and exploitability. Address CVEs identified by NVD enrichment.

7. **Enable Logging & Monitoring** — MEDIUM

   Enable comprehensive logging on all discovered services. Forward logs to a SIEM platform for real-time correlation and anomaly detection.

8. **Conduct Regular Security Assessments** — MEDIUM

   Schedule periodic scans using NET_SCAN to track changes in attack surface. Compare exposure scores over time to measure security improvement.

Disclaimer

This report is generated by NET_SCAN, an automated security assessment tool. The findings are based on network-level TCP port scanning and heuristic vulnerability mapping. This assessment does not constitute a full penetration test. Results should be validated by qualified security *professionals. The tool performs only passive reconnaissance and does not exploit any vulnerabilities.*